

Maxspeed for Vietnam's national road network

The bet: **maxspeed** across ~28,000 km of national roads & expressways – launching **early September 2026**.



Foundational data

An estimation problem

Human-confirmed, with evidence

12.9%

Almost the entire network **has no speed limit**

Of **133,771 km** of tertiary+ roads in OSM, only **12.9%** carry a speed limit – **~116,000 km missing**.

This is the biggest gap for **routing, ETA and safety** – the foundation to compete as a national map platform (vs VietMap LIVE) and serve **VETC's 4.2M vehicles**.

Maxspeed for ~28,000 km of national roads + expressways, in time for an early-September launch

Unit of work = a named national route / expressway (motorway · trunk · primary).
Minor roads dropped; central Hanoi skipped (already done).

SCOPE

National roads & expressways

motorway / trunk / primary – the most-driven roads, with the most consistent limits.

PRINCIPLE

Evidence mandatory

Every edit carries a street-view image in the changeset. Copying without evidence → reverted + account banned.

METHOD

Manual → SOP → AI

Tuệ + Quân screen-record to build an SOP, train ~20–30 pay-per-km editors, then automate.

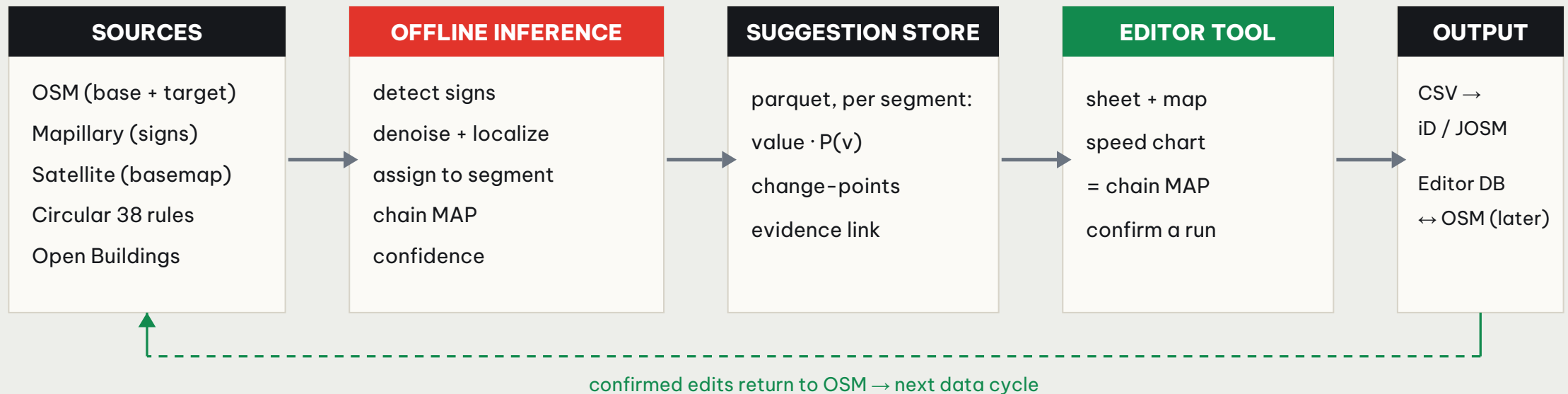
Not 28,000 km of manual work – one estimation problem

The real question: **what is the limit on each stretch, and how sure are we?** Then spend editor-hours only where the math says a human is needed.

$$P(\text{speed} \mid \text{evidence}) \propto$$
$$P(\text{speed} \mid \text{road morphology}) \leftarrow \text{prior: legal default (Circular 38)}$$
$$\times \prod P(\text{observation} \mid \text{speed}) \leftarrow \text{independent evidence: street-view signs}$$
$$\times \prod P(\text{speed} \mid \text{neighbour seg}) \leftarrow \text{coupling: speed is spatially autocorrelated}$$

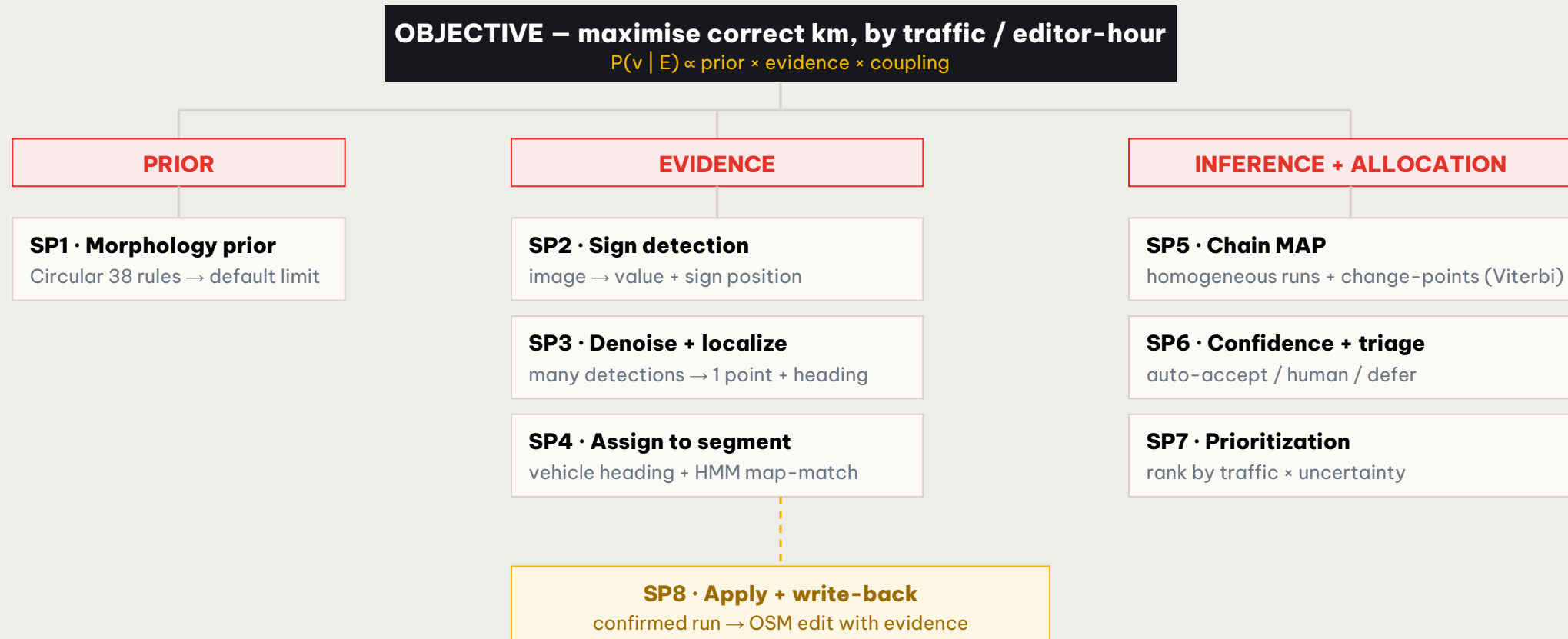
AI detects · Rules + graph reason · Humans confirm – the architecture throughout (RoadTagger, He et al. 2020).

Inference runs **offline**; the browser only renders suggestions



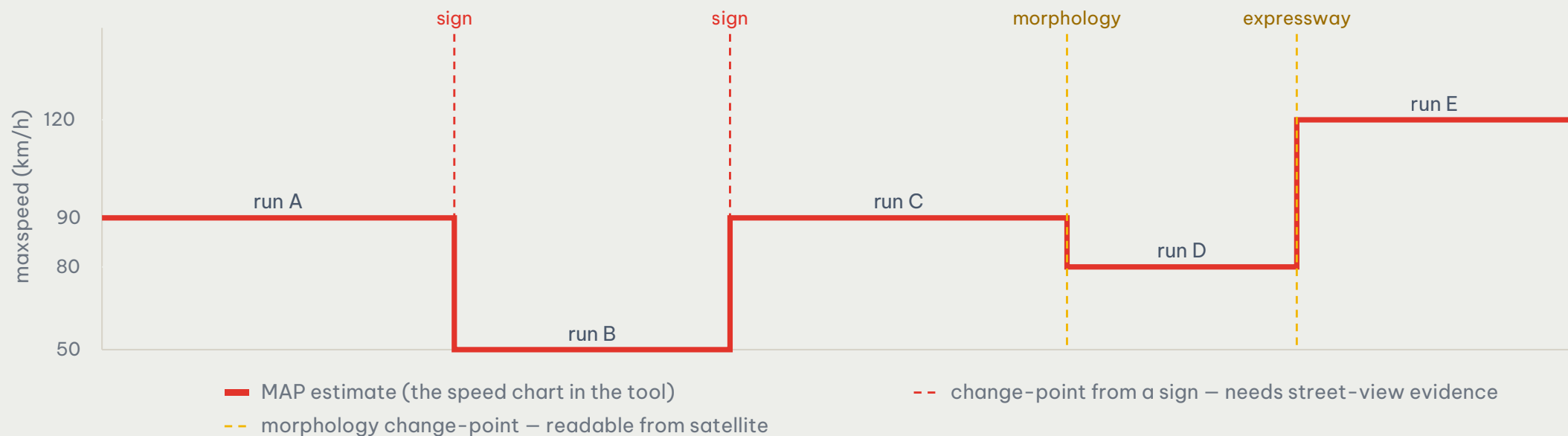
One objective → **eight sub-problems**, each built & tested alone

KM 05
8 sub-problems



Confirm the **change-points**, not every little segment

The limit holds over long **homogeneous runs**, jumping only at change-points: a sign, a built-up boundary, a morphology change, a major junction.

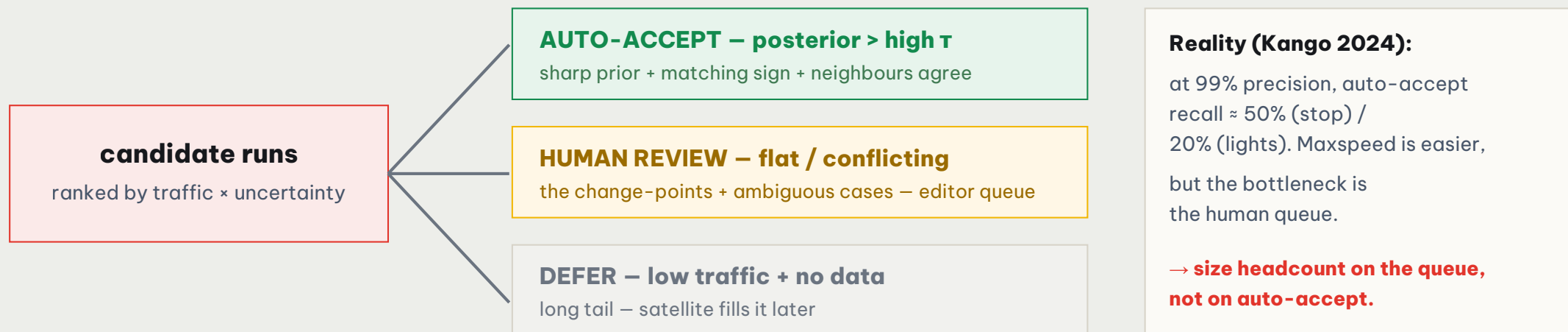


National roads & expressways – every factor favours them

FACTOR	ON THE BIG ROADS	EFFECT
Prior $P(v \text{morphology})$	uniform + well-tagged morphology (expressway→120, divided rural→90)	sharp prior, low ambiguity
Chain coupling	long homogeneous runs, few change-points / 100 km	one observation propagates over many km
Evidence cost	driven most → dense Mapillary	cheap to confirm
Value	highest traffic	each edit worth the most

Dense **Mapillary** is both the traffic proxy (value) and the evidence-availability proxy (cost) – and they coincide on exactly the roads we care about. Note: it's a biased proxy (contributor behaviour, not AADT) – fine for ranking, not a ground-truth volume.

The posterior shows **where humans are actually needed**



Circular 38 keys the limit on **morphology, not road class**

SATELLITE

Morphology → default prior

divided/single, lane count, built-up. **At launch: free basemap only** (read by eye). Purchased + CV later.

STREET-VIEW

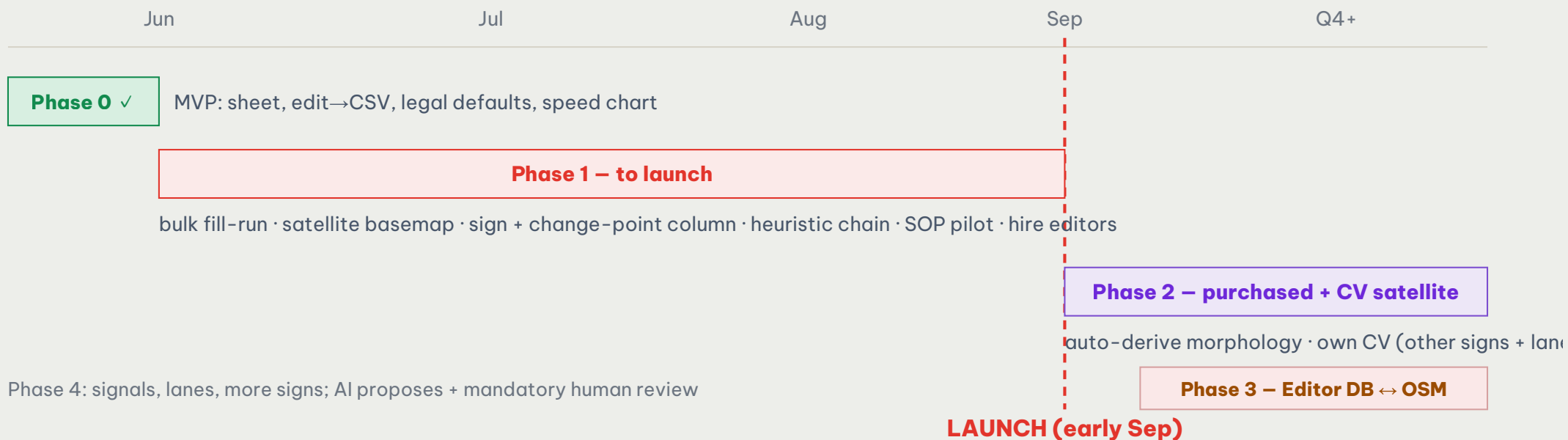
Signs → override the default

these are the change-points. **At launch: the primary evidence** (Mapillary → own dashcam).

Compliance line (do not blur): satellite → legal-default limit is legitimate, scalable evidence. A **posted** (non-default) limit still needs the sign image. Cross this line → changesets reverted + accounts banned.

Only **Phase 1** is on the launch critical path

KM 10
To
September



The same editor becomes a multi-tool QA platform

KM 11

One editor,
many tools

Leadership flagged five OSM tools to fold in. The lazy-strategic move: **don't embed five sites – give the editor a tool suite, like Bolt's Atlas**. One map shell + one data substrate, each tool a lens. The shell is traffic/lanes today; the substrate is the planned Editor Spatial DB. New capabilities plug in as modules, reusing the worklist + evidence policy.

MODULE	RUNS ON	WHEN
maxspeed · lanes · signs	live OSM (Overpass)	now
Turn restrictions · max-height / truck	live OSM (Overpass) – zero infra	Phase 2
Change-tracking – who edited what, with evidence	Editor Spatial DB (diff feed)	Phase 3
Parking + capacity	Editor Spatial DB (osm2pgsql)	Phase 3+

Highest-leverage tool: change-tracking. With 20–30 paid-per-km editors and evidence-per-edit mandatory, a "what changed in this corridor, by whom, with what evidence" feed is direct QA – and it's exactly what the Editor DB is for, so it justifies that backend rather than just consuming it. Bolt Atlas itself isn't integratable (private GPS) – we copy the checks, not the site.

Five decisions that **belong to leadership**

- **Evidence policy** (sets the rollout speed ceiling). Recommended: **hybrid** – posted (non-default) limits need street-view; visible-morphology defaults tagged `source=survey/imagery` at human pace, not a bot import.
- **Imagery source & budget** – internal tile server vs commercial sub-metre; how much, where. **Start procurement in parallel now** so it lands after launch.
- **Build vs buy CV** – own model (owns evidence + retraining loop) vs vendor.
- **Write-back timing** – how long to stay export-first before funding the Editor Spatial DB.
- **Editor operations** – headcount, pay-per-km rate, QA gate.

Low-hanging fruit first. **Measure. Scale.**

An estimation problem, not 28,000 km of manual work. Confirm the change-points on the busiest roads, auto-fill the rest – enough to launch maxspeed in early September.

Approve the evidence policy

Kick off imagery procurement

Approve editor headcount